Living provisional claims

Source SHA-256: aabd9cba683f4170719852cfe173cc9f4fdf8b05cd231f56711b1136295705a6
Status: provisional research register; no entry establishes methodological novelty.
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Research novelty and depth register (D-26)

This is a living register, not a novelty claim. It connects each potential contribution to executable evidence and records what still needs a closest precedent and corpus recheck. Claim maturity is intentionally conservative: every entry is not established methodological novelty, and the development fixture remains `not_assessed`.

Potential contribution	Executable evidence	Closest precedent / current basis	Unresolved search	Claim maturity
Canonical policy composition for binary CSS BP	`src/bp_registry/canonical/contracts.py`, `execution.py`, seven-role manifests, `tests/canonical/test_contracts.py`	Existing BP implementations commonly fix update/schedule choices; this project’s evidence is a procedure contract, not a claim of first abstraction	Map decoder-policy frameworks and QLDPC BP variants for comparable explicit role/lifecycle schemes	implementation-backed design; provisional
Typed work, budgets, lineage, and independent evaluation boundary	`budgets.py`, `lineage.py`, `execution.py`, work/lineage tests and canonical structural ledger	Work accounting and provenance are established concerns; exact closest QLDPC decoder evidence register remains incomplete	Search QLDPC reproduction/provenance and work-normalization practice	implementation-backed design; provisional
Local profiling with representative pairing	`profiling.py`, `tests/canonical/test_profiling.py`, D-20 strategy	Google-Wide Profiling and stratified-sampling profiling motivate overhead and sampling separation	Recheck decoder-benchmark methodology and whether an equivalent clean/profile semantic-pair contract exists	potential contribution; provisional pending corpus recheck
Selective tracing with content-addressed sidecars	`tracing.py`, `tests/canonical/test_tracing.py`, D-20 strategy	Event tracing and content hashes are established engineering patterns; this project has no verified closest decoder-specific selective-trace precedent yet	Search decoder debugging/tracing, replay, and rare-event evidence practices	potential contribution; provisional pending corpus recheck
Memory-policy effects across binary CSS and joint quaternary Pauli BP	Planned Phase 10 kernel-aware policy derivations and Y/correlation fixtures; planned Phase 11 matched-sample, matched-budget memory/relay ablations; no implementation or measurements yet	[Müller et al., Relay-BP](https://arxiv.org/html/2506.01779) studies circuit-level XZ/XYZ decoding using binary error-mode matrices and priors. Its XYZ treatment is not itself a four-state I/X/Y/Z Pauli BP memory rule. This distinction is an inference from the paper's stated factorization, not an established novelty claim.	Search quaternary Pauli BP with damping, momentum, iteration memory, and relay transfer; derive source-compatible four-state equations and compare closest precedents before claiming a new method	research question only; no implementation, comparison, or novelty established

Cross-kernel memory question for later phases

The proposed study asks whether a **specified memory or relay mechanism** changes logical outcome, convergence, and work in the same way under separate binary CSS inference and joint quaternary Pauli inference on matched physical samples. Phase 4 establishes and probes memory policies in the binary CSS kernel. Phase 10 must derive and verify any mathematically compatible four-state counterpart; Phase 11 owns matched joint-versus-separate code-capacity comparisons. Phase 13 addresses Relay-BP's original circuit setting. A shared policy name is not evidence that the binary scalar recurrence, quaternary vector update, and circuit error-mode graph have the same semantics.

The [Relay-BP paper](https://arxiv.org/html/2506.01779) defines its circuit decoding problem with binary error-mode variables, a check matrix, an action matrix, and independent mode priors after graph construction. Its XZ and XYZ circuit decoding strategies differ in which binary decoding objects they use. **Inference:** even when XYZ decoding retains cross-type circuit information, it does not constitute a quaternary Pauli BP kernel with a joint I/X/Y/Z variable posterior. A quaternary memory

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extension therefore needs its own equation, normalization and state-transfer semantics, compatibility proof or explicit adaptation label, and independent Y/correlation fixtures.

Before promoting a claim, the project must identify the closest published quaternary memory and relay methods, prespecify matched samples and budgets, retain negative and inconclusive results, and distinguish a mathematical policy extension from an empirical observation about its effect. This entry tracks a candidate research contribution; it does not assert priority or improved performance.

How to use the register

- **Potential contribution** states the narrow method or artifact, not a superlative. It must be traceable to executable evidence and tests.
- **Closest precedent** records a cited or explicitly incomplete comparison. “No verified closest precedent yet” means unresolved search, not novelty.
- **Unresolved search** is carried forward into the mapping study and must be refreshed before an abstract, manuscript, or review asserts differentiation.
- **Claim maturity** may advance only after corpus recheck, precise comparison, and evidence appropriate to the proposed claim. Development output is not assessed reproduction, population performance, or established methodological novelty.

Locked evidence limitations

Profiling and selective tracing are deliberately listed as provisional. A representative paired cohort can support a specified profiling estimate only when selection precedes outcomes, inclusion probability is retained, and semantic agreement is established. Targeted diagnostic replays and bounded traces remain explanatory evidence, not population evidence. The canonical three-qubit development result is canonical implementation evidence, not a source-faithful reproduction or a statistical reproduction claim.

Before manuscript use, re-run the unresolved search against the expanded literature corpus, record the exact compared implementations and definitions, and downgrade or remove any statement that lacks a defensible closest precedent.